Project Sentinel — Iteration 2

Richer features (SOFA + nephrotoxins) + Colab retrain: discrete difference vs Iteration 1

Report date 2026-07-21 · Stage-2 LightGBM @ 24h · same 359,603-row / 4,480-stay test split

One-line result: adding 8 physiologically-motivated features (197 → 205) and retraining changed 24h AUROC by **+0.001 (0.885 → 0.886)** — flat, as predicted. But one new feature, **hours-since-nephrotoxin, is now the #5 most important feature**, so the additions are informative, not useless; their signal simply overlaps what creatinine already provides.

1. What was done this iteration

- **Resolved a Colab merge.** Pulled the retrained models + notebook run-outputs from Colab and merged with the local EHR-integration work. Two whole-file notebook conflicts (01, 06) were resolved by taking the Colab side — the pipeline code was identical on both, so only run-outputs differed. The *"new architecture"* commit was a misnomer: it changed only 11 lines of `fhir_source.py`, not the model.
- **Added richer features (197 → 205).** Partial SOFA (4 of 6 organ systems: coag/liver/cardio/renal) and nephrotoxic-drug exposure (NSAIDs + aminoglycosides from prescriptions → active flag + hours-since).
- **Retrained on Colab** (30k-stay full-MIMIC sample, same split) and re-evaluated on the held-out test set.

2. Discrete metric difference (Stage-2 LightGBM @ 24h)

Metric	Iter 1 (197 feat)	Iter 2 (205 feat)	Δ
Features	197	205	+8
AUROC @6h	0.934	0.935	+0.001
AUROC @12h	0.902	0.903	+0.001
AUROC @24h	0.885	0.886	+0.001
AUPRC @24h	0.770	0.772	+0.002
Accuracy @24h	0.821	0.823	+0.002
Precision @24h	0.582	0.588	+0.006
Recall @24h	0.765	0.760	–0.005
F1 @24h	0.661	0.663	+0.002
Conformal abstention	17.2%	16.4%	–0.8pp

Every discrimination metric is within ±0.006 of Iteration 1. The model traded a hair of recall for a hair of precision (slightly fewer false alarms, slightly more misses). Nothing regressed; nothing jumped.

3. Confusion matrix shift (24h, threshold 0.5)

Cell	Iter 1	Iter 2	Δ
True negatives (TN)	232,212	233,601	+1,389
False positives (FP)	45,219	43,830	–1,389 (fewer false alarms)
False negatives (FN)	19,320	19,709	+389 (more missed AKI)
True positives (TP)	62,852	62,463	–389

Interpretation: at threshold 0.5 the new model is marginally more conservative. For an early-warning use the small recall drop matters more than the precision gain — but it is tiny, and the operating threshold (lower it toward 0.35) is the real lever, not the feature set.

4. Subgroup fairness (unchanged, still good)

Subgroup	Iter 1 AUROC	Iter 2 AUROC
Adult (<65)	0.894	0.895
Elderly (65+)	0.873	0.874
Gap	0.021	0.021

5. Did the new features actually get used?

Yes — and this is the most informative result. Ranked by LightGBM gain importance among all 205 features:

New feature	Importance rank (of 205)	Read
hours_since_nephrotoxin	5	Top-tier — recency of nephrotoxic exposure is a strong signal
nephrotoxin_active	118	Modest
sofa_partial	124	Modest — the composite organ score
sofa_liver / sofa_renal	140 / 142	Low (overlap creatinine/bilirubin already present)
sofa_coag / sofa_cardio	172 / 191	Low
nephrotoxin_started	194	Low (the raw per-hour flag; superseded by hours-since)

Why AUROC stayed flat despite a top-5 feature: the new signals are *correlated* with features the model already had. SOFA sub-scores are computed from creatinine, MAP, platelets and bilirubin — all already present — so they mostly re-express existing information. Nephrotoxin recency adds something genuinely new (hence rank 5), but a single strong feature among 205 shifts a 0.885 AUROC only slightly.

6. Conclusion & recommendation

- **Keep the features.** They are physiologically correct, one is highly ranked, and they cost nothing at inference. They also make the model more robust when creatinine is noisy or missing (a real district-hospital case).
- **Do not expect features to be the AUROC lever.** At this data scale the model is signal-saturated on creatinine + urine. Confirmed twice now (15-trial and this run).
- **The real next gains are elsewhere:** (a) external validation on eICU / AmsterdamUMCdb — prove it generalises to another hospital; (b) robustness to missing features for low-resource sites (OpenClinic GA hospitals rarely have the full lab panel); (c) local retraining once RMRTH data exists.

This report documents Iteration 2 only. Full pipeline, methodology, and metric glossary are in the main technical report (Project_Sentinel_Technical_Report.pdf).